**Software Test Plan (STP) — Operation Theatre Management System**

**Project:** Operation Theatre Management System (OTMS)
**Version**: 1.0
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1. Introduction

### Purpose

This document defines the test plan for the Operation Theatre Management System (OTMS) v1.0. It specifies the testing objectives, scope, strategy, resources, environment, schedule, responsibilities, risks, and test deliverables required to verify that OTMS satisfies the requirements defined in the SRS.

### Scope

Testing covers user authentication and role-based access, surgery management, surgery scheduling, theatre availability, staff availability, equipment availability, scheduling conflict detection, emergency surgery prioritization, rescheduling of affected surgeries, notifications, resource management, utilization reporting, audit logging, performance, reliability, and security.

Hardware maintenance, physical operation of surgical equipment, and internal implementation of external hospital systems are excluded from the OTMS test scope.

### References

- OTMS Software Requirements Specification (SRS) v1.0
- OTMS Software Architecture and Design Specification (SAD) v1.0
- OTMS Requirements Traceability Matrix (RTM)
- Hospital Information System/API specifications
- Applicable hospital security and data-protection policies

### Definitions

- OTMS: Operation Theatre Management System

- OT: Operation Theatre
- SRS: Software Requirements Specification
- SAD: Software Architecture and Design Specification
- STP: Software Test Plan
- RTM: Requirements Traceability Matrix
- RBAC: Role-Based Access Control
- UAT: User Acceptance Testing
- FR: Functional Requirement
- NFR: Non-Functional Requirement
- TC: Test Case

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**2. Test Items**

The following OTMS modules and services will be tested:

- Authentication & RBAC
- Surgery Management
- Scheduling Engine
- Scheduling & Conflict Detection Service
- Resource Management Service
- Emergency Prioritization Service
- Notification Service
- Utilization & Reporting
- Audit & Monitoring
- REST API Gateway
- OTMS Database
- Hospital Information System/API integration

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**3. Features to be Tested**

Testing will cover the functional, non-functional, and security requirements defined in the OTMS SRS.

**Functional Features**

- OTMS-F-001: User authentication
- OTMS-F-002: Role-based access control

- OTMS-F-003: Authentication event logging
- OTMS-F-004: Create surgery record
- OTMS-F-005: Update surgery details
- OTMS-F-006: View surgery details/status
- OTMS-F-007: Schedule surgery
- OTMS-F-008: Check theatre availability
- OTMS-F-009: Prevent overlapping surgeries
- OTMS-F-010: Check doctor availability
- OTMS-F-011: Check nurse availability
- OTMS-F-012: Check equipment availability
- OTMS-F-013: Detect scheduling conflicts
- OTMS-F-014: Display conflict details
- OTMS-F-015: Mark surgery as emergency
- OTMS-F-016: Prioritize emergency surgeries
- OTMS-F-017: Identify affected surgeries
- OTMS-F-018: Reschedule affected surgeries
- OTMS-F-019: Send schedule notifications
- OTMS-F-020: Notify users affected by emergency scheduling
- OTMS-F-021: Update theatre/resource availability
- OTMS-F-022: Update equipment maintenance/availability status
- OTMS-F-023: Generate OT utilization reports

**Non-Functional Features**

- OTMS-NF-001: Scheduling/resource response time
- OTMS-NF-002: System availability
- OTMS-NF-003: Protection of sensitive information
- OTMS-NF-004: Auditability
- OTMS-NF-005: Usability
- OTMS-NF-006: Concurrent access and data integrity

**Security Features**

- OTMS-SR-001: TLS 1.2+ communication
- OTMS-SR-002: Role-based access control
- OTMS-SR-003: Security and scheduling audit logs
- OTMS-SR-004: Protection against unauthorized data access
- OTMS-SR-005: API request validation and authorization
- OTMS-SR-006: Protection of credentials and sensitive information in logs

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## 4. Features Not to be Tested

The following items are outside the OTMS test scope:

- Internal implementation of the Hospital Information System.
- Internal implementation of external notification gateways.
- Physical maintenance of operation theatre hardware.
- Physical operation or firmware of surgical equipment.
- Internal infrastructure managed by external service providers.
- Clinical decisions made by doctors or hospital staff.

External systems will be tested only to the extent required to verify OTMS integration and expected API behavior.

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## 5. Test Approach /

### Strategy Unit Testing

Individual modules and services will be tested independently.

Examples:

- Authentication logic
- Scheduling logic
- Conflict detection
- Emergency prioritization
- Resource availability checks
- Notification logic

### Integration Testing

Interactions between OTMS components and external systems will be tested.

Examples:

- OTMS UI ↔ REST API Gateway
- Scheduling Engine ↔ Resource Management Service
- Scheduling & Conflict Detection Service ↔ OTMS Database
- Emergency Prioritization Service ↔ Scheduling Service

- OTMS ↔ Hospital Information System/API
- OTMS ↔ Notification Gateway

### System Testing

Complete end-to-end workflows will be tested.

Examples:

- Normal surgery scheduling
- Scheduling with unavailable resources
- Scheduling conflict detection
- Emergency surgery scheduling
- Rescheduling affected surgeries
- Notification of affected users

### Acceptance Testing

Hospital stakeholders will verify that the system satisfies operational requirements.

Primary acceptance users include:

- OT Coordinator
- Doctor / Surgeon
- Nurse
- OT Technician
- Hospital Administrator
- Hospital Management

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## Testing Types

### Functional Testing

Verify that all functional requirements operate according to the SRS.

### Regression Testing

Verify that changes to scheduling, emergency prioritization, resources, or notifications do not break previously working functionality.

### Performance Testing

Verify scheduling and resource availability response times under expected concurrent usage.

### Usability Testing

Verify that users can understand and operate the OTMS interface efficiently.

### Security Testing

Verify authentication, RBAC, authorization, secure communication, audit logging, and protection of sensitive information.

### Integration Testing

Verify correct communication with the Hospital Information System/API and notification services.

### Concurrency Testing

Verify that simultaneous scheduling operations do not create inconsistent or conflicting database states.

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## Entry Criteria

Testing may begin when:

- A stable OTMS build is available.
- Required test cases have been prepared.
- Test data is available.
- Test environment is configured.
- Required APIs and integrations are available.
- Critical blocking development issues have been resolved.

## Exit Criteria

Testing may be completed when:

- 100% of planned test cases have been executed.
- All critical and high-severity defects are resolved or formally accepted.
- Required functional requirements have passed testing.
- Security objectives and security requirements have been validated.
- Performance acceptance criteria have been satisfied.
- UAT acceptance criteria have been satisfied.

### 5.1 Security Validation

Security testing will include:

- Authentication validation.
- Role-based access-control testing.
- Unauthorized-access testing.
- API authorization testing.
- TLS 1.2+ verification.
- Input validation and malicious-input testing.
- Audit-log verification.
- Verification that passwords and sensitive information are not stored in plaintext logs.
- Verification that restricted surgery/patient information cannot be accessed by unauthorized roles.
- Security testing of emergency scheduling and schedule modification operations.

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## 6. Test

### Environment

### Hardware

- Developer/test workstations
- Hospital workstation simulation
- Application server
- Database server
- Network infrastructure
- Optional notification device/environment

OTMS does not directly control surgical equipment; therefore, equipment availability will be represented through test data or system interfaces.

### Software

- OTMS application v1.0
- OTMS Database
- REST API Gateway
- Hospital Information System/API sandbox

- Notification service/gateway sandbox
- Supported web browser
- Operating system supporting the application

**Testing Tools**

- Postman: API and integration testing
- JMeter: Performance and load testing
- Selenium: UI automation
- Jira: Defect tracking
- Draw.io: UML/design documentation where required

**Test Data**

Test data will include:

- Test user accounts for each OTMS role
- Valid and invalid authentication credentials
- Surgery records
- Emergency surgery records
- Theatre schedules
- Doctor availability
- Nurse availability
- Equipment availability
- Conflicting surgery schedules
- Alternative theatre/time-slot data
- Equipment maintenance records
- Notification recipients
- Utilization data

All patient-related test data should be dummy/non-production data.

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**7. Test Schedule**

| Milestone | Planned Date |
|---|---|
| Test Plan preparation | 07-Oct-2026 |

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|---|---|
| Test case design | 08-Oct-2026 – 10-Oct-2026 |
| Test environment setup | |
| Test data preparation | |
| Test execution start | |
| Functional testing | |
| Integration testing | |
| Security testing | |
| Performance testing | |
| Regression testing | |
| UAT | |
| Test execution completion | |
| Test Summary Report | |

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**8. Test Deliverables**

The following deliverables will be produced:

- Software Test Plan
- Test Case Specification
- Manual Test Cases
- Automated Test Scripts
- API Test Collection
- Test Data

- Test Execution Logs
- Defect Reports
- Requirements Traceability Matrix
- Performance Test Results
- Security Test Results
- UAT Results
- Test Summary Report

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**9. Roles and Responsibilities**

| Role | Name | Responsibility |
|---|---|---|
| QA Lead | Sudhiksha | Prepare test plan, coordinate testing, monitor test progress and approve test completion |
| Test Engineer | Suravi | Design and execute test cases, record results and report defects |
| Developer | Suravi | Fix defects, support debugging and participate in defect triage |
| OT Coordinator / Domain Representative | Praneetha | Validate scheduling workflows and operational requirements |
| Hospital Administrator | Shesanth | Validate administrative and resource-management functionality |
| Product Owner | Shesanth | Review test results and approve release readiness |

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**10. Risks and Mitigation**

| Risk | Mitigation |
|---|---|

| | |
|---|---|
| Delay in stable OTMS build delivery | Request early builds and perform incremental testing |
| Test environment unavailable | Maintain a backup test environment and database |
| Hospital Information System/API unavailable | Use a controlled API sandbox/mock service |
| Incorrect or incomplete test data | Prepare and validate test data before execution |
| Scheduling logic changes during testing | Maintain regression test suite and update affected test cases |
| Emergency scheduling creates complex conflicts | Prepare dedicated conflict and rescheduling scenarios |
| Notification service unavailable | Use notification gateway stubs/mocks during functional testing |
| Concurrent scheduling causes inconsistent data | Perform concurrency and transaction-integrity testing |
| Security defects discovered late | Perform security validation throughout development rather than only at the end |

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## 11. Assumptions & Dependencies

- OTMS SRS and SAD remain the baseline documents for testing.
- Required test users and role assignments will be available.
- Test environment will contain representative theatre, staff, equipment, and surgery data.
- Hospital Information System/API test environment will be available.
- Notification gateway test environment will be available.
- Test data will not contain real patient information.
- Required development builds will be delivered according to the project schedule.
- Testers will have appropriate access to OTMS modules and test environments.

## 12. Suspension & Resumption

**Criteria Suspend testing if:**

- Test environment is unavailable for more than 4 hours.
- A build contains blocking defects preventing more than 30% of planned test cases from being executed.
- Database or test data becomes unreliable.
- Required external APIs are unavailable for an extended period.
- A critical security issue prevents safe continuation of testing.

**Resume testing if:**

- Blocking defects have been resolved or formally accepted.
- Test environment has been stabilized.
- Required test data has been restored.
- External dependencies are available.
- A stable build has been delivered.

## 13. Test Case Management & Traceability

The Requirements Traceability Matrix (RTM) will be used to ensure that every important SRS requirement is mapped to one or more test cases.

Examples:

| SRS Requirement | Test Case(s) |
|---|---|
| OTMS-F-001 — Authentication | TC-AUTH-01, TC-AUTH-02 |
| OTMS-F-002 — RBAC | TC-AUTH-03, TC-AUTH-04 |
| OTMS-F-004 — Create Surgery Record | TC-SURG-01 |
| OTMS-F-007 — Schedule Surgery | TC-SCH-01 |

| | |
|---|---|
| OTMS-F-008 — Theatre Availability | TC-SCH-02 |
| OTMS-F-009 — Prevent Overlapping Surgeries | TC-SCH-03 |
| OTMS-F-010 — Doctor Availability | TC-RES-01 |
| OTMS-F-011 — Nurse Availability | TC-RES-02 |
| OTMS-F-012 — Equipment Availability | TC-RES-03 |
| OTMS-F-013 — Conflict Detection | TC-CON-01 |
| OTMS-F-015 — Emergency Surgery | TC-EMG-01 |
| OTMS-F-016 — Emergency Prioritization | TC-EMG-02 |
| OTMS-F-017 — Identify Affected Surgeries | TC-EMG-03 |
| OTMS-F-018 — Reschedule Affected Surgery | TC-EMG-04 |
| OTMS-F-019 — Schedule Notifications | TC-NOT-01 |
| OTMS-F-020 — Emergency Notifications | TC-NOT-02 |
| OTMS-F-023 — Utilization Reports | TC-REP-01 |
| OTMS-NF-001 — Response Time | TC-PERF-01 |
| OTMS-NF-002 — Availability | TC-REL-01 |
| OTMS-SR-001 — TLS 1.2+ | TC-SEC-01 |

| | |
|---|---|
| OTMS-SR-002 — RBAC | TC-SEC-02 |

| | |
|---|---|
| OTMS-SR-003 — Audit Logging | TC-SEC-03 |
| OTMS-SR-004 — Unauthorized Access | TC-SEC-04 |
| OTMS-SR-005 — API Authorization | TC-SEC-05 |
| OTMS-SR-006 — Sensitive Data in Logs | TC-SEC-06 |

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**14. Test Metrics & Reporting**

The following metrics will be collected during testing:

- Total test cases
- Test cases executed
- Test cases passed
- Test cases failed
- Test cases blocked
- Test execution percentage
- Pass percentage
- Requirement coverage
- Defect count
- Defect severity
- Defect density
- Defect aging
- Number of reopened defects
- Regression test pass rate
- Performance response time
- Security test pass rate

**Reports**

The following reports will be generated:

- Daily Test Execution Report

- Defect Status Report
- Requirements Coverage Report
- Performance Test Report
- Security Test Report
- UAT Report
- Final Test Summary Report

**15. Approvals**

| Role | Name | Signature / Date |
|---|---|---|
| QA Lead | Suravi | |
| Development Lead | Sudhiksha | |
| OT Coordinator / Domain Representative | Praneetha | |
| Product Owner | Shesanth | |